An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

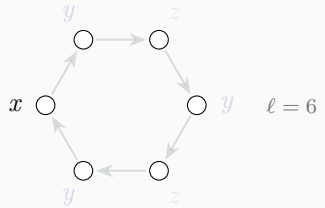
Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

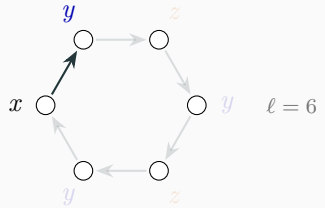
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

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- 辺を 1 本ずつ出し、式の対応する部分と一緒に読む
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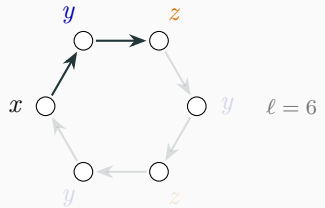
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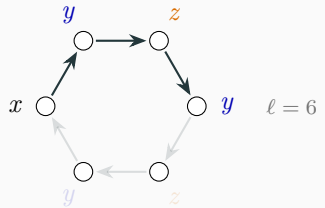
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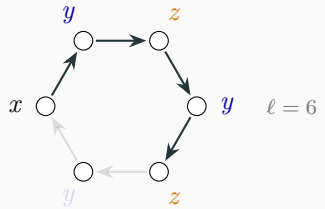
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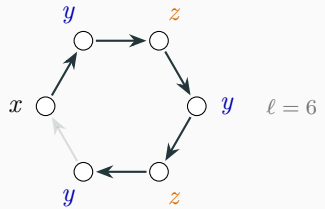
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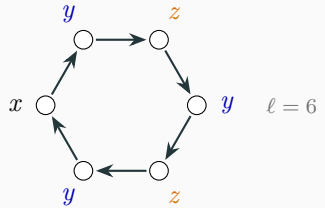
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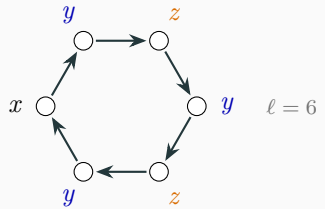
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FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
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CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow 2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

スライド概要：$k \geq 4$ では n^k が最適、FO^3 だけ n^ω に下がる。代償は翻訳の $2^{O(|\varphi|)}$。

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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

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(In this translation, we use 2^k compositions)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS]**)**

Can we prove that there is no polynomial-size translation from FO^3 *to* CoR*?*

Partial answer: Yes for $\Sigma_1\mathrm{CoR}$ (with transitive closure) [Nak22, JLAMP],
which has the same expressive power as **existential** FO^3, which is closed under **extensions**.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

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Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ — before: exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]; $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with transitive closure X^*

👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
- 閉包付き CoR：$2 \cdot \#\text{合成} + \#\text{閉包} \geq 2^k$
- 行列言語：Brijder–Gyssens–Van den Bussche の問いに答える
- 変わるのは「1 演算が気づける変更の数」だけ（量化子 1、集約 3β、閉包 1）
- 論文には他に：CoR 等式、逆なし回路（ちょうど 2^k）、乱択回路、ランダム入力

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スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

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Thm. (This paper)

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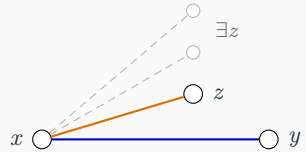
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \big[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z))\big] \qquad (|G_k| = O(k))$$



at every i: one of the edges xy, xz is in P_i,
the other is in Q_i

DNF: $G_k \equiv \bigcup_{A \subseteq [k]} \big[H_A \cap (H_{A^c}\,;\mathbf{1})\big]$: 2^k **compositions** $(H_A \coloneqq \bigcap_{i\in A} P_i \cap \bigcap_{i\notin A} Q_i)$

$\Rightarrow$ optimal up to a factor of 2: every equivalent CoR term needs $\geq 2^{k-1}$ compositions

スライド概要：G_k は DNF で合成 2^k 個、これは 2 倍を除いて最適。

メモ：

- 各 i で、辺 xy と辺 xz の一方が P_i、他方が Q_i
- 下界は合成以外の演算（補・逆・共有）を無料にしても成り立つ

Outline

1. Our results

2. Technical overview

3. Conclusion

Proof idea: telling $\mathcal{M}_0$ apart from 2^k structures

Goal $C \equiv G_k \implies C$ has $\geq 2^{k-1}$ compositions (C: any CoR term or circuit)

C must tell $\mathcal{M}_0$ apart from each of 2^k structures $\mathcal{M}_\lambda$: $G_k = 1$ on $\mathcal{M}_0$, $G_k = 0$ on $\mathcal{M}_\lambda$

$\mathcal{M}_0$

CoR operation	meaning	newly tells $\mathcal{M}_0$ apart from
$\mathbf{0}, \mathbf{1}, \mathbf{id}$, $\cup, \cap, \neg$, $X^\smile$	constants, Boolean operations, converse	no $\mathcal{M}_\lambda$
$X \,;\, Y$	composition = Boolean matrix product	≤ 2 of the $\mathcal{M}_\lambda$

every $\mathcal{M}_\lambda$ needs a composition that tells it apart $\implies$ $2^k \leq 2 \cdot \#\text{compositions}(C)$

スライド概要：C は $\mathcal{M}_0$ と 2^k 個の $\mathcal{M}_\lambda$ を見分けねばならない。合成 1 個が新たに見分けられるのは高々 2 個。

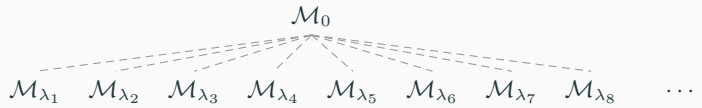
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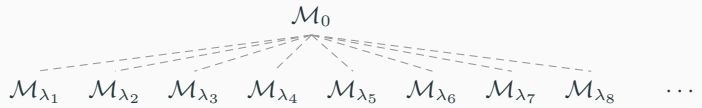
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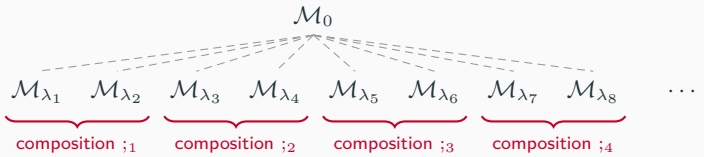
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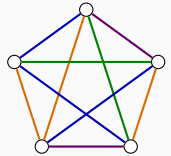
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The structure $\mathcal{M}_0$



an edge-colored complete graph

colors $\mu \subseteq [k]$; on an edge of color μ: $P_i \iff i \in \mu$, $Q_i \iff i \notin \mu$

$\Rightarrow$ if xy has color μ: $G_k(x, y)$ holds iff some edge xz has color μ^c

for all $u \neq v$ and all colors μ, ν: at least **two** middle vertices z

(random coloring, $n \geq n_0(2^k)$ [JMT95, Algebra Univers.])

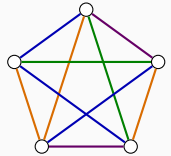
fix such an $\mathcal{M}_0$ and one vertex $\hbar$, the **hub**

スライド概要：$\mathcal{M}_0$ は辺に色の付いた完全グラフ。どの色の組にも中間の頂点が 2 つ。hub を 1 つ固定。

メモ：

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- 「中間の頂点が 2 つ」が最後のページで効く

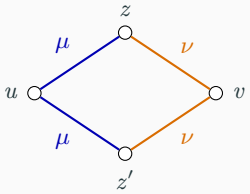
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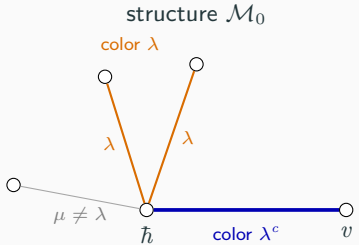
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Recoloring at the hub $\hbar$



$G_k(\hbar, v) = 1$

$G_k(\hbar, v)$ needs an edge $\hbar z$ of color λ

recoloring λ changes **exactly those edges at** $\hbar$, and nothing else

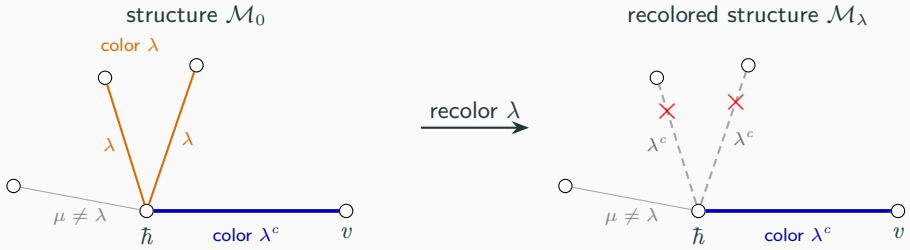
2^k recolored structures $\mathcal{M}_\lambda$, one for each color λ

スライド概要：塗り替えで G_k が 1 から 0 に変わる。

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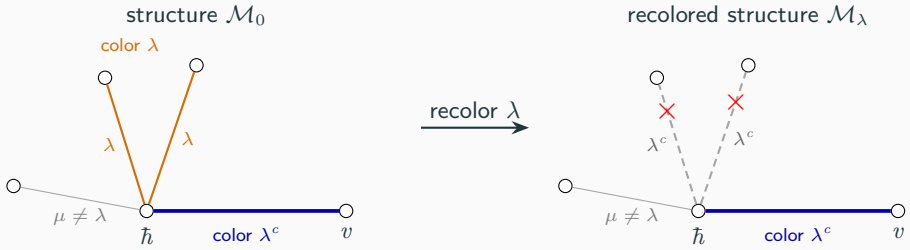
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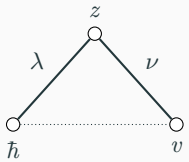
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Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
$\Rightarrow$ each composition detects at most two colors λ

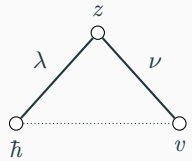
スライド概要：$\hbar$ を含む点対では気づかれうる、含まない点対では気づかれない。

メモ：

- $\hbar$ を含まない点対では、$\hbar$ を通らない中間の頂点 z' が残る
- 条件は 3 つ、どれも λ を決める $\rightarrow$ 高々 2

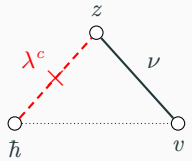
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(b) $\mathcal{M}_\lambda$, at $(\hbar, v)$



every middle vertex is lost:
detected

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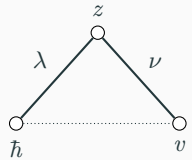
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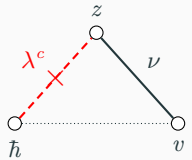
Why a composition detects at most two recolorings

(a) $\mathcal{M}_0$, at $(\hbar, v)$



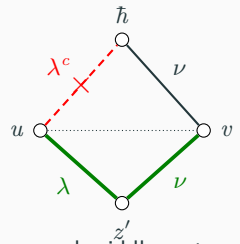
middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

(b) $\mathcal{M}_\lambda$, at $(\hbar, v)$



every middle vertex is lost:
detected

(c) $\mathcal{M}_\lambda$, at (u, v)



a second middle vertex z'
avoids $\hbar$: not detected

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ
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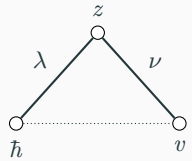
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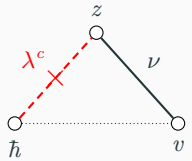
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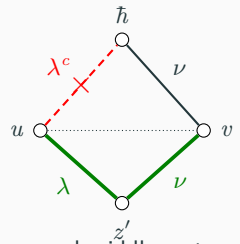
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Outline

1. Our results

2. Technical overview

3. Conclusion

Summary

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- some FO^3 formulas of size $O(k)$ need $\geq 2^{k-1}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k recolorings of one structure $\mathcal{M}_0$; each composition detects ≤ 2 of them.

? Open questions

- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
- the exact number of compositions for G_k: between 2^{k-1} and 2^k?

Thank you for your attention!

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メモ：

- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

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